

CHARTERHALL, United Kingdom

WMO: 031580

Lat: 55.7086N

Lon: 2.3847W

Elev: 364

StdP: 14.50

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	28.0	22.0	16.7	30.5	24.0	18.4	32.0	36.7	47.2	33.2	46.0	5.5	290	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	73.7	63.8	70.1	61.9	67.4	60.2	65.6	71.4	63.4	68.4	61.5	65.9	9.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	88.8	67.8	61.2	82.0	65.7	59.5	76.8	63.8	30.6	71.7	28.9	68.6	27.6	66.0	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.9	25.0	21.8	DB	20.4	79.9	4.3	3.8	17.3	82.7	14.7	84.9	12.3	87.1	9.2	89.9
			WB	21.1	70.5	3.1	4.1	18.9	73.4	17.0	75.9	15.3	78.2	13.0	81.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	38.3	39.3	41.9	45.4	49.7	54.7	58.5	58.2	54.6	49.0	42.8	38.7	
	DBStd	8.68	4.93	5.04	4.85	4.76	4.79	4.10	4.28	4.00	3.89	4.82	5.17	6.02	
	HDD50	1791	363	300	254	151	65	6	0	0	5	76	221	350	
	HDD65	6348	828	719	716	588	476	310	209	214	312	497	665	814	
	CDD50	930	0	1	3	13	54	146	262	256	144	45	5	1	
	CDD65	11	0	0	0	0	0	0	6	4	1	0	0	0	
	CDH74	100	0	0	0	0	0	5	52	40	3	0	0	0	
	CDH80	10	0	0	0	0	0	0	4	6	0	0	0	0	
Wind	WSAvg	9.7	11.0	11.8	11.4	10.0	9.3	8.6	7.7	8.2	9.1	9.8	9.8	10.4	
Precipitation	PrecAvg	34.0	3.2	2.4	2.2	2.4	2.4	2.7	2.8	3.0	2.6	3.5	3.7	3.3	
	PrecMax	44.2	7.2	5.2	4.4	5.7	4.1	6.5	6.0	6.4	5.2	7.2	6.7	7.1	
	PrecMin	25.3	1.0	0.6	0.9	0.6	0.7	0.6	1.1	0.6	0.9	1.1	1.3	1.3	
	PrecStd	4.8	1.3	1.2	0.9	1.4	0.9	1.5	1.4	1.5	1.2	1.5	1.4	1.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.8	54.1	58.9	65.0	69.9	74.1	79.6	79.3	72.0	63.7	58.0	54.3	
		MCWB	49.9	50.1	51.3	54.4	60.3	63.3	65.2	66.9	63.1	58.2	54.5	51.9	
	2%	DB	50.2	51.3	54.5	59.3	65.4	69.3	74.5	72.3	67.7	60.6	54.8	51.7	
		MCWB	47.7	48.1	48.5	51.9	57.7	61.1	64.4	63.4	60.2	56.6	52.2	49.5	
	5%	DB	48.1	49.1	51.9	56.1	61.4	66.2	70.5	68.7	64.5	58.5	52.8	49.9	
		MCWB	45.7	46.3	47.3	50.0	54.6	58.8	62.8	61.3	58.2	55.0	50.2	47.6	
	10%	DB	46.2	47.4	49.6	53.6	58.4	63.2	67.1	66.1	62.1	56.5	51.0	47.7	
		MCWB	43.9	44.5	45.8	48.4	52.6	56.9	60.6	59.5	57.0	53.4	48.6	45.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	51.0	52.5	56.1	63.3	66.0	69.7	68.5	65.4	60.1	54.9	52.3
			MCDB	52.3	53.5	56.8	62.6	68.0	71.3	73.5	72.7	69.9	62.1	57.0	53.9
2%		WB	48.1	48.6	49.9	52.9	58.4	62.3	66.0	64.7	61.8	57.4	52.6	49.9	
		MCDB	50.1	50.9	53.3	58.1	63.8	67.4	72.9	70.3	65.3	59.6	54.5	51.7	
5%		WB	46.0	46.8	48.0	51.0	55.6	59.8	63.6	62.6	59.5	55.6	50.7	48.0	
		MCDB	47.8	48.9	51.1	55.3	60.3	64.8	68.9	67.6	63.1	58.0	52.4	49.8	
10%		WB	44.2	44.9	46.3	49.1	53.3	57.7	61.6	60.8	57.6	53.8	48.9	45.9	
		MCDB	46.1	47.2	49.4	53.0	57.3	62.0	66.2	64.7	60.8	56.2	50.6	47.7	
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.6	11.6	13.2	14.3	14.2	15.5	15.6	13.8	11.5	9.8	8.7	
		MCDBR	10.4	10.9	14.6	17.3	19.3	19.7	21.2	19.2	18.2	13.0	11.2	10.8	
		MCWBR	9.4	9.0	10.2	11.1	12.5	12.6	12.8	11.4	11.9	9.8	9.8	10.1	
	5% WB	MCDBR	10.2	10.4	13.5	16.4	17.7	18.0	18.8	17.4	16.4	12.2	10.8	10.7	
		MCWBR	9.4	9.1	10.4	11.3	12.6	12.8	12.9	11.9	12.5	10.2	9.9	10.1	
Clear-Sky Solar Irradiance	taub	0.313	0.329	0.359	0.380	0.378	0.379	0.379	0.369	0.353	0.336	0.311	0.295		
	taud	2.286	2.375	2.319	2.289	2.341	2.363	2.367	2.407	2.460	2.455	2.354	2.252		
	Ebn at Noon	188	235	254	266	274	274	272	267	256	231	190	164		
	Edh at Noon	17	23	30	36	36	36	35	32	26	21	16	14		
All-Sky Solar Radiation	RadAvg	169	371	704	1149	1509	1469	1414	1145	817	447	218	124		
	RadStd	24	36	78	122	105	136	144	87	64	51	20	12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page